October 15, 2026

The Editors-in-Chief
Pattern Recognition

Dear Editors,

We submit the manuscript "Training parts versus matching parts in pose-based gait recognition" for consideration as a regular research article in *Pattern Recognition*.

Part-aware gait recognizers usually add two things at once: identity losses on body regions during training and fusion of regional similarities at retrieval. Gains are then credited to one of them without testing the other. Our study separates the two. On one fixed pose-graph encoder, seven training objectives, including global and part supervised contrastive losses and cross-entropy controls, were trained with five seeds each on CASIA-B development identities. No auxiliary objective differed reliably from a metric-learning baseline, and an apparent penalty of part supervision in an earlier three-seed experiment did not replicate. In contrast, fusing normalized raw part pools from frozen checkpoints improved corrupted-probe accuracy under every objective. On Gait3D, with the recipe frozen on development identities and the test identities evaluated once, the same matcher raised Rank-1 from 21.88% to 23.86% and reached parity with GPGait, which we reproduced with its official configuration on identical data.

We then applied the matcher to GPGait itself, and the result reversed: GPGait's part-specific branches, trained with their own losses, add 9.1 points on CASIA-B and 5.2 on Gait3D, while raw part matching lowers its accuracy. The paper therefore concludes that where a part-aware gain comes from depends on the encoder, and that global, raw regional and trained regional matching should be compared on the same frozen checkpoints. We report the limitations openly, including a large accuracy gap to GPGait on CASIA-B, earlier project exposure to the CASIA-B test identities, an available-pose subset of Gait3D and the absence of labelled real occlusion.

We believe the paper suits *Pattern Recognition* because it addresses a methodological question about how part-aware recognition systems are evaluated, and supports its answer with controlled experiments, seed-by-identity bootstrap intervals, reproduced baselines and a transfer test on an established encoder. Every table and figure can be regenerated from the saved per-probe outcomes with the released scripts.

This manuscript is original, has not been published, and is not under consideration for publication elsewhere. Both authors approved the submission and declare no competing interests. The study received no specific funding. The use of an AI assistant during manuscript preparation is declared

in the manuscript.

Sincerely,

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